



## Data Article

# BanglaDial: A merged and imbalanced text dataset for Bengali regional dialect analysis

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## ABSTRACT

BanglaDial is a unified corpus of Bengali dialectal text, consisting of 60,729 sentence-level entries representing eleven regional dialects and Standard Bangla. It reflects phonological, lexical, and syntactic divergences among dialects, offering a comprehensive linguistic spectrum beyond Standard Bangla. The dataset was compiled from online public repositories, representing regions such as Chittagong, Sylhet, Barisal, Rangpur, and others. All samples are written in Bengali script and annotated with their respective dialect labels, capturing key phonological, lexical, and syntactic variations across dialects. Bengali is the world's seventh most spoken language, with over 265 million speakers, and is known for its vast literary heritage and linguistic complexity. However, existing NLP (Natural Language Processing) technologies are predominantly trained on Standard Bangla, leaving dialectal varieties significantly underrepresented and poorly supported in digital tools. BanglaDial addresses this gap by providing a curated and annotated resource to support automatic dialect identification and the development of dialect-aware natural language processing applications. This dataset also contributes to broader goals of linguistic preservation and digital inclusion for underrepresented Bengali-speaking communities.

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Specifications Table

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subject                | Computer Sciences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Specific subject area  | Underrepresented Languages, Bangla Dialects, Low-Resource Languages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Type of data           | Table (text/string)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| How data were acquired | The dataset is created by combining subsets of 4 corpora.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Data collection?       | BanglaDial is created in two steps: 1) Data sources identification and 2) data preparation and insertion. Four publicly available corpora were identified, analyzed and filtered to build BanglaDial described in section data descriptions. Each corpus supports a set of multiple dialects.                                                                                                                                                                                                                                                                                                                                                            |
| Data source location   | The list of the primary data sources used to create BanglaDial dataset is as follows.<br>Primary data sources:<br>Vashantor [4]: <a href="https://data.mendeley.com/datasets/bj5jgk878b/2">https://data.mendeley.com/datasets/bj5jgk878b/2</a><br>ONUBAD [5]: <a href="https://data.mendeley.com/datasets/6ft99kf89b/2">https://data.mendeley.com/datasets/6ft99kf89b/2</a><br>Bhashamul [6]: <a href="https://www.kaggle.com/competitions/regipa/data">https://www.kaggle.com/competitions/regipa/data</a><br>Bangla Dialect Dataset [7]: <a href="https://data.mendeley.com/datasets/sm63ryv5dt/1">https://data.mendeley.com/datasets/sm63ryv5dt/1</a> |
| Data accessibility     | Repository name: Mendeley Data<br>Data identification number: <a href="https://doi.org/10.17632/sx6ybcps2n.2">10.17632/sx6ybcps2n.2</a><br>Direct URL to data:<br><a href="#">BanglaDial: A Merged and Imbalanced text Dataset for Bengali Regional dialect analysis.</a><br>- Mendeley Data                                                                                                                                                                                                                                                                                                                                                             |

1. Value of the Data

- BanglaDial is the first large-scale Bengali dialectal dataset that contains 11 regional dialects alongside Standard Bengali. It includes major areas of influence of dialects such as Chittagong, Sylhet, and Rangpur and Standard Bangla. Such a range makes it a significant sample of the collection of dialects of one of the world’s most widely used languages.
- At present, many advanced generative AI models have limited exposure to low-resource languages which leaves Bengali dialects underrepresented. BanglaDial addresses this gap by serving as a foundational resource, enabling the development of translation systems, dialect identification frameworks, and multilingual chatbots capable of comprehending and producing text across these dialects. BanglaDial has practical applications in voice assistants, and educational apps that can adapt responses to a user’s dialect.
- Linguists and socio-linguists can use BanglaDial to perform corpus-based dialectometric, enabling research into phonological, lexical, and syntactic variations among regional dialects. The dataset is a useful tool for demographic analysis and geolocation in digital communication
- BanglaDial promotes diversity and inclusion in language technology by prioritizing the preservation and respect of Bengali dialects. It is also informative, with potential to provoke research and awareness about low-resource dialects in South Asia.
- The data exhibits imbalance due to the natural distribution of dialectal samples collected from various sources. This renders BanglaDial highly valuable for research in imbalanced data learning, e.g., research on resampling algorithms, cost-sensitive learning, data augmentation, and fairness-aware modelling for NLP.

2. Background

Bengali or Bangla is the seventh most spoken language in the world with close to 272.7 million native speakers in Bangladesh, parts of India, and a broad diaspora [1]. Although Bangla

is widely used across formal, academic, and official contexts, natural language processing efforts have largely concentrated on Standard Bangla, with limited attention to the rich variation present in regional dialects.

In other regions of Bangladesh, people speak significantly different Bengali dialects, such as Chittagonian, Sylheti, and Barisali. The dialects often differ in pronunciation, grammar, and vocabulary, hence being difficult for Standard Bangla speakers to understand. Around 13 million people speak the Chittagonian dialect, 11.8 million speak Sylheti, and around 549,031 people speak the Barisali dialect in southern Bangladesh. Besides this, the diaspora Sylheti community in the UK includes around 400,000 speakers [2]. Such dialectic difference is a major impediment to communication and translation, particularly in online platforms, educational content, and public services. Therefore, the development of translation models, dialect identification systems, and language tools capable of processing regional Bangla dialects becomes increasingly important [3].

To meet these needs, the BanglaDial dataset was built. It is a merged and cleaned corpus derived from four public datasets of dialectal text samples in Bangla script: Vashantor [4], ONUBAD [5], Bhashamul (REGIPA) [6], and the Bangla Dialect Dataset [7]. Collectively, all these datasets cover twelve dialect regions in Bangladesh. Merging was done by identifying dialect-labelled corpora, cleaning texts, and placing them into a uniform format.

BanglaDial offers a vital resource to build inclusive NLP systems, improve dialectal translation acc, and enrich understanding of dialectal variation. Furthermore, because it is imbalanced in terms of its class distribution, it is also well-suited to research on data augmentation, resampling, and cost-sensitive learning that are vital for real-world language modelling.

### 3. Data Description

The objective of BanglaDial is to build a diverse and comprehensive dataset that captures a wide range of Bengali regional dialects, thereby supporting better generalization for dialect classification models. Our dataset design and collection methodology were inspired by the IADD dataset [8], which presents a large-scale corpus for Arabic dialect identification across multiple regions. Similarly, BanglaDial aims to capture dialectal diversity across Bengali-speaking regions in Bangladesh. BanglaDial is created in two main steps:

1. Data sources identification, and
2. Data preparation and insertion.

At the end of the process, BanglaDial is stored in a CSV format with the following key fields:

- **Sentence:** Contains the dialectal text sample written in Bengali script.
- **Language:** Indicates the corresponding dialect label used for classification.

We have made our datasets openly accessible on a data repository [9]. There are 60,729 sentences in the dataset class wise distribution in Fig. 1.

Table 1 presents a detailed breakdown of sentence counts and their corresponding percentages for each dialect. Additionally, Fig. 2 visualizes the origin of the dataset by mapping sentence contributions based on data sources. To further enrich the linguistic analysis, Table 2 shows the average word count, total word count, and average character count for each dialect, offering insights into the structural variations and verbosity across regional speech patterns within the dataset.

Table 3 provides a representative sample of the BanglaDial dataset, showcasing example sentences from each of the 12 regional dialects alongside their corresponding labels. The dataset is formatted as a CSV file, where each entry consists of two columns: the sentence and its associated dialectal class. This structure allows for straightforward parsing and label-based processing in natural language tasks. To illustrate phonetic and lexical differences, we have added transliteration, the standard Bengali and English equivalent for each example. The table demonstrates the

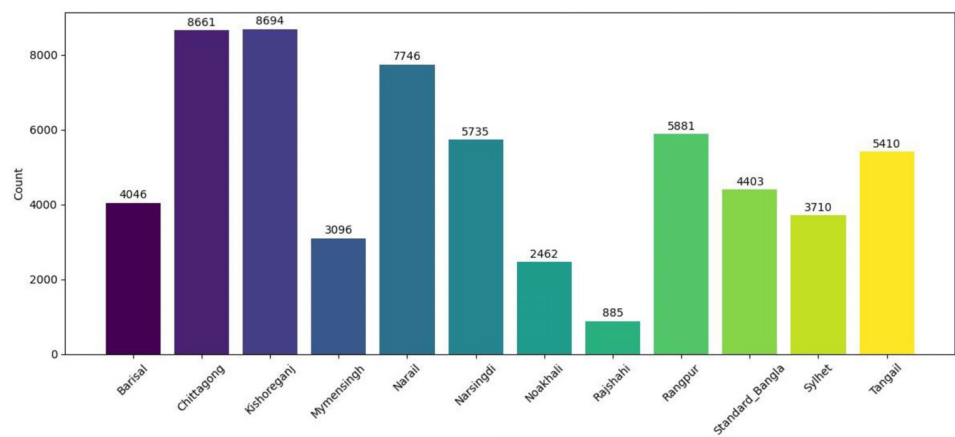


Fig. 1. Dataset class wise description.

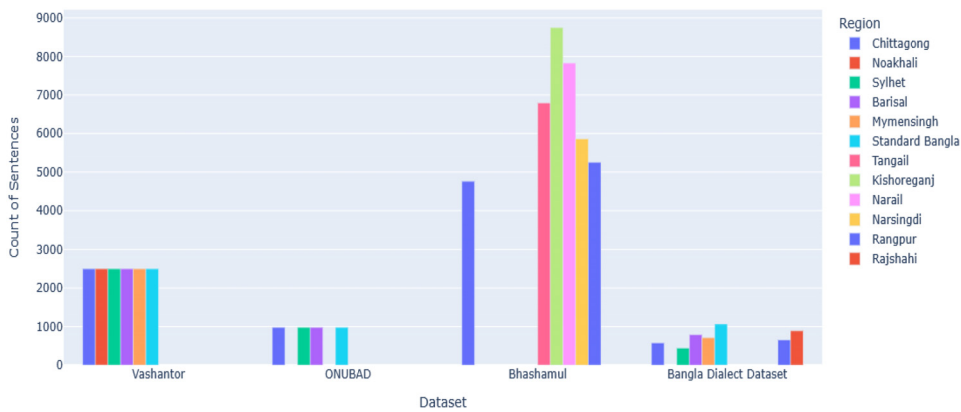


Fig. 2. Sample size by region and by data source.

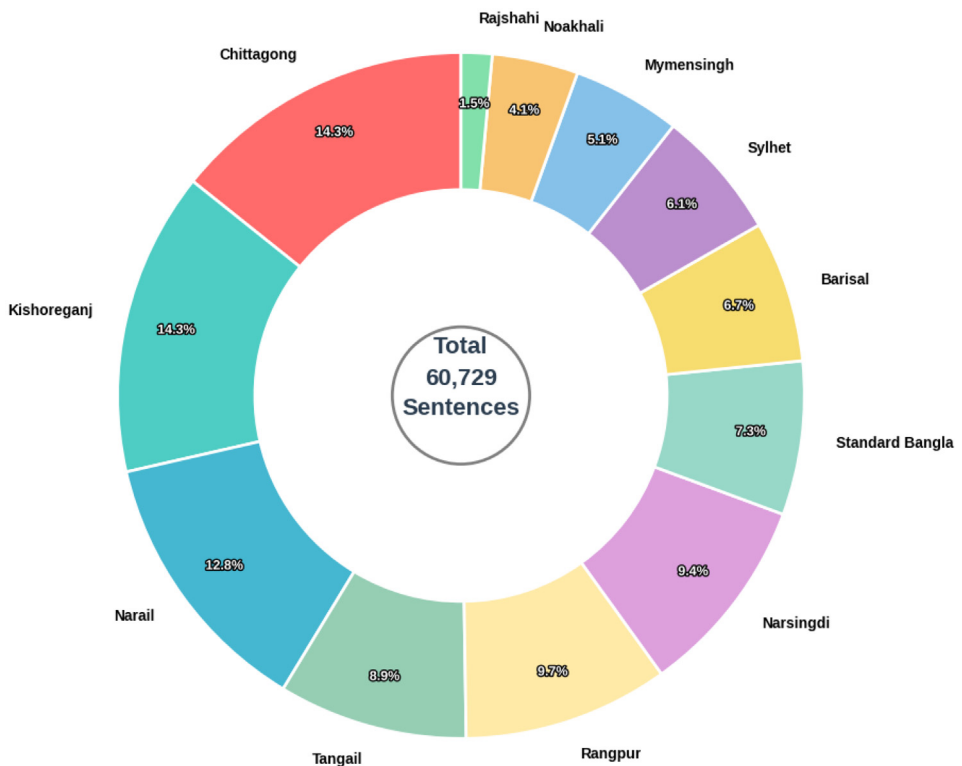
Table 1  
Detailed overview of BanglaDial.

| SL | Language        | Sentences | Percentage   |
|----|-----------------|-----------|--------------|
| 1  | Chittagong      | 8661      | 14.26 %      |
| 2  | Kishoreganj     | 8694      | 14.31 %      |
| 3  | Narail          | 7746      | 12.76 %      |
| 4  | Tangail         | 5410      | 8.90 %       |
| 5  | Rangpur         | 5881      | 9.68 %       |
| 6  | Narsingdi       | 5735      | 9.44 %       |
| 7  | Standard_Bangla | 4403      | 7.25 %       |
| 8  | Barisal         | 4046      | 6.66 %       |
| 9  | Sylhet          | 3710      | 6.11 %       |
| 10 | Mymensingh      | 3096      | 5.09 %       |
| 11 | Noakhali        | 2462      | 4.05 %       |
| 12 | Rajshahi        | 885       | 1.46 %       |
|    | <b>Total</b>    | 60,729    | <b>100 %</b> |

**Table 2**

Data description.

| Region          | Word Count | Avg Word Count | Avg Char  |
|-----------------|------------|----------------|-----------|
| Chittagong      | 60,045     | 6.932802       | 36.148135 |
| Kishoreganj     | 43,832     | 5.041638       | 25.629285 |
| Narail          | 45,111     | 5.823780       | 28.756907 |
| Tangail         | 31,557     | 5.833087       | 29.679482 |
| Rangpur         | 35,391     | 6.017854       | 29.746642 |
| Narsingdi       | 34,531     | 6.021099       | 30.950131 |
| Standard Bangla | 25,348     | 5.756984       | 29.849421 |
| Barisal         | 23,476     | 5.802274       | 31.632229 |
| Sylhet          | 22,361     | 6.027224       | 31.009973 |
| Mymensingh      | 19,644     | 6.344961       | 33.809432 |
| Noakhali        | 16,387     | 6.655971       | 34.500406 |
| Rajshahi        | 4747       | 5.363842       | 29.526554 |
| Total           | 362,430    |                |           |

**Fig. 3.** Proportional distribution class wise.

phonetic, lexical, and syntactic variations that distinguish one dialect from another. The comparison with related dataset is given in [Table 4](#).

[Fig. 3](#) and [4](#) present the dialect distribution and the average sentence length distribution respectively. [Table 5](#) shows the overall imbalance indicators where class distribution exhibits moderate long-tail imbalance: the top-5 classes account for 60.5 % of all sentences (top-10 94.5 %); Shannon entropy [12]  $H = 3.416H=3.416$  (normalized 0.9530.9530.953); imbalance ratio IR [13]=9.82.

**Table 3**  
Example of the dataset.

| Sentence                                                             | Transliteration                                                              | Standard Bangla                                                                         | English                                                                                                             | Dialects               |
|----------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------|
| আই জানি এডে বউত মাইফোয়া কমেণ্ট ফরিবার লাই আইবো                      | Āi jani ede bouth maifoṯa koment foribar lai aibo                            | আমি জানি এখানে অনেক ছেলে-মেয়ে মন্তব্য পড়তে আসবে                                       | I know many boys and girls will come here to read the comments.                                                     | <b>Chittagong</b>      |
| এলাই তো আন্দের সর্বনাশ কইরা দিছে                                     | Elai to anner sorbonash koira diche                                          | এরাই তো আপনার সর্বনাশ করে দিয়েছে                                                       | They are the ones who have ruined you.                                                                              | <b>Kishoreganj</b>     |
| দিপু কাকা আমারে ধরিছে বিটা যাতি অবেনে কাইলকে                         | Dipu kaka amare dhoriche bita jati abene kailke                              | দিপু কাকা আমাকে ধরেছেন, কাল সেখানে যেতে হবে                                             | Uncle Dipu caught me, I have to go there tomorrow.                                                                  | <b>Narail</b>          |
| বাই, আমনোগো এনু কি ইকটু ফুন চার্জ দ্যান যাবো?                        | Bai, amnego enu ki ektu pun charj dyan jabo?                                 | ভাই, আপনাদের এখানে কি একটু ফোন চার্জ দেয়া যাবে?                                        | Brother, can I get a little phone charge here from you?                                                             | <b>Tangail</b>         |
| আর ফিরুজ হইচে উয়ারা বাপের মতানা                                     | Ar Firoz hoiche uyar baper moton                                             | আর ফিরোজ হয়েছে ওর বাবার মতো                                                            | And Firoz has become like his father.                                                                               | <b>Rangpur</b>         |
| ঘরের ভিত্তে গিয়াও হেই হমানে চিল্লান শুরু করছে                       | Ghorer vitte giyao hei homane chillan shuru korse                            | ঘরে ভিতরে গিয়েও সেইভাবে চিৎকার শুরু করেছে                                              | As soon as he went inside the room, he started shouting in that way.                                                | <b>Narsingdi</b>       |
| আজকে আমার মন ভালো নেই                                                | Ajke amar mon bhalo nei                                                      | আজ আমার মন ভালো নেই                                                                     | Today I am not in a good mood.                                                                                      | <b>Standard Bangla</b> |
| আব্বা মোর লইজা লাল জামা কিন্যা আনছে                                  | Abba mor loiggya lal jama kinya anshe                                        | বাবা আমার জন্য লাল জামা কিনে এনেছে                                                      | Father bought me a red shirt.                                                                                       | <b>Barisal</b>         |
| আমি আফনার লাগি কিতা করতাম ফারি                                       | Ami afnar lagi kita kortam fari                                              | আমি আপনার জন্য কী করতে পারি?                                                            | What can I do for you?                                                                                              | <b>Sylhet</b>          |
| আপনে কি এই ব্যাফারে কিসু কইতারবাইন                                   | Apne ki ei byafare kisu koitarbain                                           | আপনি কি এই বিষয়ে কিছু বলতে পারবেন?                                                     | Can you say something about this matter?                                                                            | <b>Mymensingh</b>      |
| ফোন হালাই লেপটিনে গেলে মনে অয় লাজেন জরুরী ফাইল হালাই অফিসে বসি রইছি | Phone halai leptine gele mone oi lajen joruri file halai office boshi roichi | যখন আমি আমার ফোনটি রেখে টয়লেটে যাই, তখন মনে হয় যেন একটা জরুরি ফাইল রেখে অফিসে বসে আছি | When I leave my phone and go to the toilet, it feels like I am sitting in the office leaving behind an urgent file. | <b>Noakhali</b>        |
| আমার চা ভালাগে, আমার বুনের আবার কফি ভালাগে                           | Amar cha bhal lage, amar buner abar coffee bhal lage                         | আমার চা ভালো লাগে, আমার বোনের আবার কফি ভালো লাগে                                        | I like tea, but my sister likes coffee.                                                                             | <b>Rajshahi</b>        |

**Table 4**

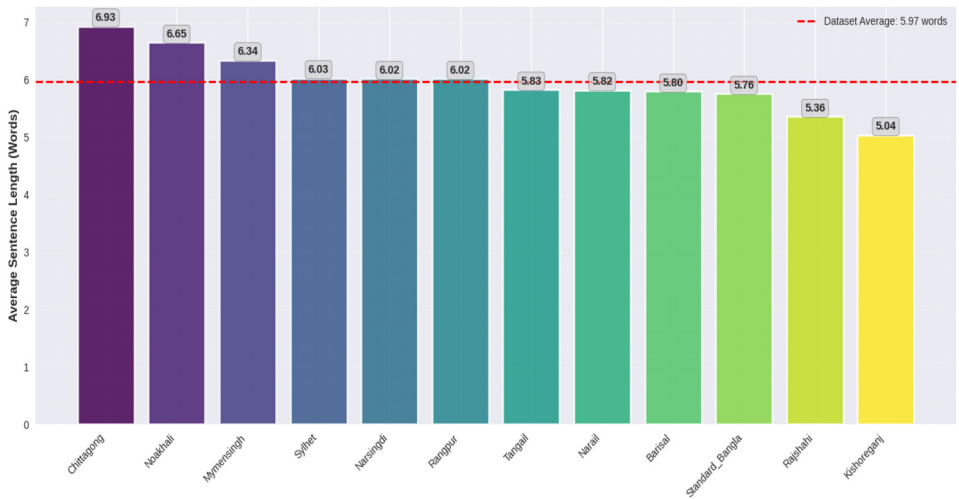
Comparison with previous datasets.

| Corpus                             | Data Source Type | Sentence | Number of Dialects | Dialects Language |
|------------------------------------|------------------|----------|--------------------|-------------------|
| IADD [8]                           | Secondary        | 135,804  | 6                  | Arabic            |
| IIT Bombay Eng–Hindi Corpus [10]   | Secondary        | 1492,827 | 2                  | English, Hindi    |
| Arabic Dialect Identification [11] | Secondary        | ~2.9M    | 5                  | Arabic            |
| Our Dataset BanglaDial [9]         | Secondary        | 60,729   | 12                 | Bangla            |

**Table 5**

Imbalance indicators.

| Indicator                         | Value      |
|-----------------------------------|------------|
| Top-3 cumulative proportion       | 41.3 %     |
| Top-5 cumulative proportion       | 60.5 %     |
| Top-10 cumulative proportion      | 94.5 %     |
| Shannon entropy (H, base-2)       | 3.416 bits |
| Normalized entropy $H / \log_2 C$ | 0.953      |
| Imbalance ratio (IR = max/min)    | 9.82       |

**Fig. 4.** Average sentence length distribution by class.

## 4. Experimental Design, Materials and Methods

### 4.1. Methodology

#### 4.1.1. Data sources identification

BanglaDial is constructed by combining subsets of four publicly available corpora: Vashantor, ONUBAD, Bhashamul, and the Bangla Dialect Dataset. The corpora consists of samples taken from varied collections of Bengali regional dialects, as listed in Table 6.

The Vashantor corpus [4] includes dialectal Bengali sentences from various regions with a focus on lexical and phonological variation. ONUBAD [5] comprises a collection of translated texts of regional language usage in different socio-cultural contexts. Bhashamul [6], a competition dataset for dialect identification, comprises labeled data in a number of dialect zones focusing on classification tasks. Bangla Dialect Dataset [7] supplements dialectal differences collected from native Bangladeshis.

**Table 6**  
Description of corpora composing our datasets.

| Corpus                     | Source   | Dialects (Regional Level)                                                       | Sentence | Number of Dialects |
|----------------------------|----------|---------------------------------------------------------------------------------|----------|--------------------|
| Vashantor [4]              | Mendeley | Chittagong, Noakhali, Sylhet, Barisal, Mymensingh and Standard Bangla           | 15,000   | 6                  |
| ONUBAD [5]                 | Mendeley | Barisal, Sylhet, Chittagong and Standard Bangla                                 | 3920     | 4                  |
| Bhashamul [6]              | Kaggle   | Tangail, Kishoreganj, Narail, Narsingdi, Rangpur, and Chittagong                | 39,152   | 6                  |
| Bangla Dialect Dataset [7] | Mendeley | Rajshahi, Sylhet, Chittagong, Standard Bangla, Rangpur, Mymensingh, and Barisal | 5131     | 7                  |

These resources were thoroughly reviewed, filtered, and combined to create a coherent yet imbalanced dataset, aimed at advancing research in dialect recognition, low-resource language processing, and computational sociolinguistics for Bengali.

4.1.2. Data preparation and insertion

Data preparation and integration from individual sources into the BanglaDial dataset are outlined below.

4.1.3. Vashantor

Sentences from the Vashantor corpus were sourced from Mendeley Data [4]. This dataset contains regional dialects from Chittagong, Noakhali, Sylhet, Barisal, Mymensingh, and Standard Bangla, totaling 6 dialect classes. The dataset is in CSV format and includes manually curated translations. Sentences were directly inserted into BanglaDial, with each instance labeled according to its regional tag. No major preprocessing was required beyond standard normalization and cleaning.

4.1.4. ONUBAD

The ONUBAD dataset [5], also from Mendeley Data, provides translated dialectal sentences from Barisal, Sylhet, Chittagong, and Standard Bangla, spanning 4 dialect categories. Data entries are structured similarly to Vashantor. Sentences were directly added into BanglaDial, maintaining their original dialect labels. ONUBAD sentences required basic preprocessing, including the removal of punctuation and stop words, to align with BanglaDial’s formatting standards.

4.1.5. Bhashamul

The Bhashamul dataset [6] was obtained from the REGIPA competition on Kaggle. It contains 6 dialect regions: Tangail, Kishoreganj, Narail, Narsingdi, Rangpur, and Chittagong. Sentences in this dataset were annotated at the utterance level and exported in CSV format. Only instances with clear dialect tags were retained. All entries were mapped to the appropriate region label before inclusion in BanglaDial. Minor inconsistencies in dialect spelling were normalized during preprocessing.

4.1.6. Bangla dialect dataset

The Bangla Dialect Dataset [7] is the primary compilation resource for BanglaDial. It includes 7 dialects: Rajshahi, Sylhet, Chittagong, Standard Bangla, Rangpur, Mymensingh, and Barisal. This dataset integrates both crowd-sourced and institutional contributions and follows a structured CSV format with columns for sentence and dialect class. Sentences from this corpus were directly ingested into BanglaDial with minimal transformation, aside from character normalization and filtering of Latin script or noisy data.

4.1.7. Data pre-processing

After collecting and merging the dataset, the dataset went through the following pre-processing and filtering stages-



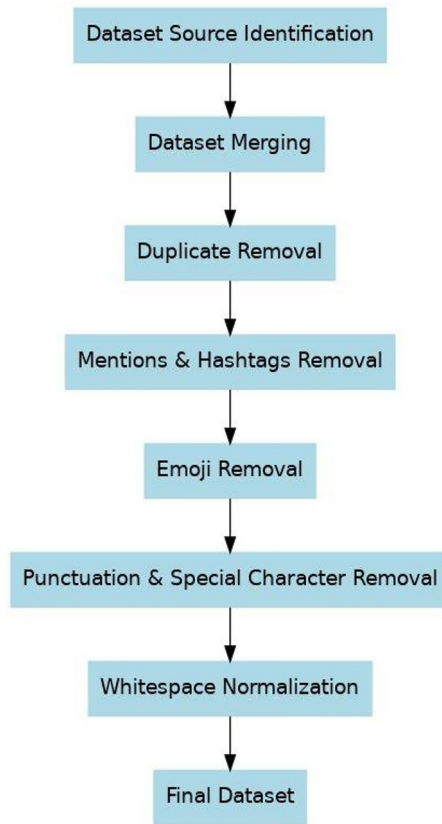


Fig. 5. Flow diagram.

### 1. Duplicate Removal

After merging the dataset, using `pandas.drop_duplicates()` all duplicate sentences were identified and removed. A total of 2574 duplicate sentences ( $\approx 4.1\%$  of all records). If the same sentence appeared more than once within or across files, only the first unique occurrence was kept.

### 2. Mentions and Hashtags Removal

Social-media style text often contains user mentions (e.g., @username) and hashtags (e.g., #topic). These were removed with regular expressions (`@\w+` and `#\w+`).

### 3. Emoji Removal

Emoji symbols were identified through their Unicode ranges and removed from the sentences. While emojis often carry emotional meaning, they are not consistent linguistic markers of dialects, so they were excluded to keep the dataset text-focused.

### 4. Punctuation and Special Character Removal

Punctuation marks and other special symbols were removed, while retaining all Unicode letters, numbers. Sentences containing Latin-script contaminations were detected through Unicode-based regex filters.

### 5. Whitespace Normalization

Multiple spaces, unnecessary line breaks, and leading or trailing spaces were normalized using regex (`\s+`).

## 6. Final Dataset

After all preprocessing steps, the dataset contained two standardized columns: sentence (cleaned Bangla text) and label (the original region name derived from the dataset file). The final dataset was saved in both CSV and XLSX formats. After cleaning and filtering dataset the sentences remained from 63,303 to 60,729. The overall workflow is illustrated in Fig. 5. All related tables, figures, and supplementary materials are available at: <https://github.com/Mehraj-Hossain-Mahi/BanglaDial>.

## Limitations

BanglaDial is a valuable resource for Bengali dialectal analysis, there are certain limitations:

- **Unbalanced Distribution:** The corpus reflects natural imbalances in dialect usage, with certain dialects of regions (e.g., Standard Bangla, Chittagonian) being more represented than others. This can bias machine learning algorithms unless balancing methods are employed in addition.
- **Text-Only Representation:** All the samples are in text form in Bengali script. The absence of phonetic or audio data prevents detailed phonological and prosodic analysis, which is crucial for comprehensive dialect studies.
- **Source Variability:** Because the data were collected and concatenated from different online sources and corpora, inconsistency in annotation style, dialect designation, or linguistic formatting may be present.
- **No Contextual Metadata:** There is no gender, socio-economic status, or age speaker metadata available in the dataset, which would be important in demographic or sociolinguistic research.

These constraints present an opportunity for further studies to enlarge, improve, and expand the dataset for additional usefulness in low-resource NLP and dialectology.

## Ethics Statement

The authors declare that there are no ethical concerns regarding the creation of the BanglaDial dataset. All data were obtained from publicly available online sources and existing open-access corpora, in full compliance with their respective usage and redistribution policies. The dataset is fully anonymized and does not contain any personally identifiable information. No direct human subjects were involved in the data collection process, and the data used were originally collected under conditions that ensured voluntary and informed contributions.

## Credit Author Statement

**Mehraj Hossain Mahi:** Methodology, Visualization, Writing–original draft; **Anzir Rahman Khan:** Conceptualization, Validation, Writing–review & editing; **Mayen Uddin Mojumdar:** Validation, Supervision

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT for the manuscript's language improvement. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

## Data Availability

BanglaDial: A Merged and Imbalanced text Dataset for Bengali Regional dialect analysis. (Reference data) (Mendeley Data)

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The author declares that there are no known competing financial interests or personal relationships which have, or could be perceived to have, influenced the work reported in this article.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## References

- [1] M.A. Islam, M. Rahman, M.A. Hossain, Chittagonian dialect dataset for machine translation and linguistic analysis, *Int. J. Adv. Comput. Sci. Appl. (IJACSA)* 12 (4) (2021).
- [2] Bangladesh Bureau of Statistics Population and Housing Census 2011: Barisal Division, Bangladesh Bureau of Statistics, 2015.
- [3] Ethnologue, Languages of Bangladesh, in: *Ethnologue: Languages of the World*, SIL International, 2022. Available at: <https://www.ethnologue.com>
- [4] F.T. Johora, M. Bin Moin, A. Al Wase, M.R. Sani, M. Ahmmed, T. Muhammad, Vashantor: a large-scale multilingual benchmark dataset for automated translation of Bangla regional dialects to Bangla language, *Mendeley Data* 2 (2024) V, doi:10.17632/bj5jgk878b.2.
- [5] N. Sultana, R. Yasmin, B. Mallik, M.S. Uddin, ONUBAD: an extensive dataset for automated translation of Bangla regional dialects into standard Bangla language, *Mendeley Data* 2 (2024) V, doi:10.17632/6ft99kf89b.2.
- [6] REGIPA organizers, Bhashamul – A regional Bangla dialect dataset Kaggle REGIPA Competition Dataset, 2022 Available at: <https://www.kaggle.com/competitions/regipa/data>.
- [7] M.J. Naeen, S.K. Das, S.D.D. Supta, T.E. Alahi, M.F.T. Faisal, A.A. Rahat, S.J.A. Ayshe, U.M. Mahjabeen, M.T. Tahmid, M.U. Mojumdar, Bangla Dialect Dataset: exploring linguistic diversity across regions, *Mendeley Data* 1 (2025) V, doi:10.17632/sm63ryv5dt.1.
- [8] J. Zahir, IADD: an integrated arabic dialect identification dataset, *Data Br.* 40 (2022) 107777.
- [9] Mehraj Hossain Mahi, Anzir Rahman Khan, BanglaDial: a merged and imbalanced text dataset for Bengali regional dialect analysis, *Mendeley Data*, 2025 V, doi:10.17632/sx6ybcps2n.2.
- [10] A. Kunchukuttan, P. Mehta, P. Bhattacharyya, The iit bombay english-hindi parallel corpus, *arXiv preprint arXiv:1710.02855*, (2017).
- [11] O. De Gibert, G. Nail, N. Arefyev, M. Bañón, J. Van Der Linde, S. Ji, ... J. Tiedemann, A new massive multilingual dataset for high-performance language technologies, *arXiv preprint arXiv:2403.14009*, (2024).
- [12] Y. Fan, H. Huang, Data entropy-based imbalanced learning, in: *Proceedings of the Southwest Data Science Conference*, Springer, 2024, pp. 95–109.
- [13] S. Wang, L.L. Minku, X. Yao, A learning framework for online class imbalance learning, in: *Proceedings of the 2013 IEEE Symposium on Computational Intelligence and Ensemble Learning (CIEL)*, IEEE, 2013, pp. 36–45.